Summary of A Survey on Mixture of Experts

A Survey on Mixture of Experts — Summary

Overview and Motivation

This survey provides a comprehensive review of Mixture of Experts (MoE) models, addressing a significant gap in the literature, as the most recent prior review predated the "ChatGPT moment" of 2022. MoE has emerged as an effective method for scaling up model capacity with minimal computational overhead, gaining substantial traction in both academia and industry. The technique allows large language models (LLMs) to expand their parameter counts dramatically while keeping computational costs in check, adhering to scaling laws that underpin LLM evolution.

Background and Core Concept

The MoE framework rests on a simple yet powerful idea: different parts of a model, called "experts," specialize in different tasks or aspects of data. A gating network directs inputs to appropriate expert networks. In transformer-based LLMs, MoE layers typically replace feed-forward network (FFN) layers, since FFNs become computationally expensive as models scale (e.g., 90% of PaLM's 540B parameters reside in FFN layers).

The survey distinguishes two main types: - **Dense MoE**: Activates all expert networks during each iteration, yielding higher accuracy but greater computational cost. - **Sparse MoE**: Activates only a selected subset (top-k) of experts, achieving sparsity and efficiency. This introduces load balancing challenges, addressed through auxiliary loss functions promoting even token distribution.

Proposed Taxonomy

The authors introduce a novel taxonomy organizing MoE advancements across three perspectives: **algorithm design**, **system design**, and **applications**.

Algorithm Design

Gating Functions are categorized into three types: - **Sparse gating**: Activates subsets of experts, using token-choice (tokens select experts), expert-choice (experts select tokens), and non-trainable approaches. - **Dense gating**: Activates all experts, effective in LoRA-MoE fine-tuning. - **Soft gating**: Fully differentiable methods using token merging or expert merging.

Experts discussion covers network types (FFN, attention-based, CNN), hyperparameters (expert count, size, placement frequency), activation functions (ReLU to SwiGLU), and the increasingly popular **shared expert** design.

Mixture of Parameter-Efficient Experts (MoPE) combines MoE with parameter-efficient fine-tuning (PEFT) techniques like LoRA, applied to FFN, attention, transformer blocks, or every layer.

Training and Inference Schemes include Dense-to-Sparse, Sparse-to-Dense transitions, and Expert Models Merging (e.g., Branch-Train-Mix).

System Design

The survey addresses challenges from MoE's sparse, dynamic workload across three dimensions: - **Computation**: Load imbalance and overhead from gate routing, addressed by specialized GPU kernels (MegaBlocks, ScatterMoE). - **Communication**: All-to-All communication bottlenecks, mitigated through hierarchical strategies and computation-communication overlapping. - **Storage**: Memory capacity constraints addressed via selective parameter retention and offloading.

Applications

MoE has been applied across **natural language processing** (translation, question answering, code generation), **computer vision** (V-MoE), **recommender systems** (MMOE, PLE), and **multimodal applications** (LIMoE, Uni-MoE).

Challenges and Future Directions

Key open challenges include: training stability and load balancing, scalability and communication overhead, expert specialization and collaboration, sparse activation efficiency, generalization and robustness, interpretability, optimal expert architecture design, and seamless integration with existing frameworks.

Conclusion

The survey serves as an essential reference for researchers, supported by a dedicated resource repository for ongoing updates, contributing to the continued advancement of MoE technologies for increasingly complex real-world tasks.